

the health and continued operation of a remote PAFTP stack.

Per PAFTP Frame Formats, PAFTP packet receipt acknowledgements SHALL be received as unsigned 8-bit integer values in the header of a PAFTP packet. The value of this field SHALL indicate the sequence number of the acknowledged packet.

Acknowledgement of a sequence number indicates acknowledgement of the previous sequence number, if it too is unacknowledged. By induction, acknowledgement of a given packet implies acknowledgement of all packets received on the same PAFTP session prior to the acknowledged packet.

An acknowledgement is invalid if the acknowledged sequence number does not correspond to an outstanding, unacknowledged PAFTP packet sequence number. In contrast to TCP, PAFTP acks are not "free." A stand-alone ack—that is, a PAFTP packet that contains a packet receipt acknowledgement value but no buffer segment payload—consumes a slot in a remote peer's window just like any other packet. Stand-alone acknowledgement packets SHALL be acknowledged by a remote peer. The implications of this are examined in [Section 4.20.3.9, "Idle Connection State"](#).

Each peer SHALL maintain an acknowledgement-received timer. When a peer sends any PAFTP packet, it SHALL start this timer if it is not already running. The timer's duration SHALL be globally defined as PAFTP_ACK_TIMEOUT seconds, referred to as the acknowledgement timeout interval.

A peer SHALL restart its acknowledgement-received timer when a valid acknowledgement is received for any but its most recently sent unacknowledged packet. A peer SHALL stop its acknowledgement-received timer if it receives an acknowledgement for its most recently sent unacknowledged packet. If a peer's acknowledgement-received timer expires, or if a peer receives an invalid acknowledgement, the peer SHALL close the PAFTP session and report an error to the application.

Because the server's handshake response bears an implicit PAFTP sequence number of zero, a server SHALL start its acknowledgement-received timer when it sends a handshake response.

Each peer SHALL also maintain a send-acknowledgement timer. When it receives any PAFTP packet, a peer SHALL record the packet's sequence number as the corresponding PAFTP session's pending acknowledgement value and start the send-acknowledgement timer if it is not already running. The timer's duration SHALL be defined as any value less than one-half the acknowledgement timeout interval. This ensures that on a healthy [WFA-USD](#) connection, a peer will always receive acknowledgements for sent packets before its acknowledgement-received timer expires.

A peer SHALL restart its send-acknowledgement timer when a pending acknowledgement is sent, either as a stand-alone PAFTP packet or piggybacked onto an outgoing buffer segment, for any but its received unacknowledged packet with the largest packet sequence number. A peer SHALL stop its send-acknowledgement timer when a pending acknowledgement is sent, either as a stand-alone PAFTP packet or piggybacked onto an outgoing buffer segment, for its received unacknowledged packet with the largest packet sequence number. If this timer expires and the peer has a pending acknowledgement, the peer SHALL immediately send that acknowledgement, and acknowledgement number SHALL be the PAFTP packet sequence number of which all previous contiguous PAFTP packet sequence numbers are already received. If the peer sends any packet before this timer expires, it SHALL piggyback any pending acknowledgement on the transmitted packet, and acknowledgement number SHALL be the PAFTP packet sequence number of which all previous